

Growth and Decay

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write down the single multiplier for a 5% increase.

1 $100\% + 5\% = 105\%$

2 105% as a decimal is 1.05

Answer: 1.05

2 [2 marks]

£500 increases by 10%.

Work out the new amount.

1 $10\% \text{ of } 500 = 50$

2 $500 + 50 = £550$ (or $500 \times 1.1 = 550$).

Answer: £550

3 [2 marks]

A population of 8000 falls by 25%.

Work out the new population.

1 $25\% \text{ of } 8000 = 2000$

2 $8000 - 2000 = 6000$ (or $8000 \times 0.75 = 6000$).

Answer: 6000

4 [2 marks]

£1000 is invested for 2 years at 3% **simple** interest.

Work out the total amount at the end.

1 Simple interest gives the same amount each year: $3\% \text{ of } 1000 = £30$

2 Two years gives $2 \times 30 = £60$ of interest.

3 $\text{Total} = 1000 + 60 = £1060$

Answer: £1060

5

[3 marks]

£1000 is invested for 2 years at 3% **compound** interest.

(a) Work out the total amount at the end.

(b) How much more is this than with simple interest at the same rate?

1 (a) The multiplier is 1.03 each year, applied twice.

2 $1000 \times 1.03^2 = 1000 \times 1.0609 = £1060.90$

3 (b) Simple interest gives £1060 exactly.

4 $1060.90 - 1060 = £0.90$, so compound interest gives 90p more.

5 The extra comes from earning interest on the first year's interest.

Answer: (a) £1060.90 (b) 90p more

6

[3 marks]

A car is worth £12 000. Its value falls by 20% each year.

Work out its value after 2 years.

1 The multiplier for a 20% decrease is 0.8

2 $12\,000 \times 0.8 = £9600$ after one year

3 $9600 \times 0.8 = £7680$ after two years

4 (Or $12\,000 \times 0.8^2 = 12\,000 \times 0.64 = 7680$.)

Answer: £7680

7

[3 marks]

A colony of bacteria doubles in size every hour.

There are 500 bacteria at the start.

How many are there after 5 hours?

1 Doubling five times means multiplying by 2^5

2 $2^5 = 32$

3 $500 \times 32 = 16\,000$

Answer: 16 000 bacteria

8

[3 marks]

£2500 is invested at 4% compound interest per year.

Work out the value after 3 years, to the nearest penny.

1 Multiplier = 1.04, applied 3 times.

2 $1.04^3 = 1.124864$

3 $2500 \times 1.124864 = 2812.16$

Answer: £2812.16

9

[4 marks]

£5000 is invested for 3 years. Two accounts are available.

Account A: 4% simple interest per year.

Account B: 3.5% compound interest per year.

Which account gives more money after 3 years, and by how much?

1 **Account A:** 4% of 5000 = £200 each year.

2 $3 \times 200 = £600$ interest, so the total is £5600

3 **Account B:** 5000×1.035^3

4 $1.035^3 = 1.108717875$, so the total is $5000 \times 1.108717875 = £5543.59$

5 $5600 - 5543.59 = £56.41$

6 So Account A gives £56.41 more.

Answer: Account A, by £56.41

10

[4 marks]

After a 15% decrease, the value of an item is £510.

(a) Work out the original value.

(b) Work out the size of the decrease in pounds.

1 **(a)** A 15% decrease means the £510 is 85% of the original.

2 The multiplier used was 0.85, so divide to reverse it.

3 $510 \div 0.85 = £600$

4 **(b)** $600 - 510 = £90$

5 Check: 15% of 600 = £90 ✓

Answer: (a) £600 (b) £90

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Write down the single multiplier for a 12.5% decrease.

1 $100\% - 12.5\% = 87.5\%$

2 87.5% as a decimal is 0.875

Answer: 0.875

2

[2 marks]

£8000 is invested at 2% compound interest per year.

Work out the value after 5 years, to the nearest penny.

1 $1.02^5 = 1.1040808032$

2 $8000 \times 1.1040808032 = 8832.646\dots$

Answer: £8832.65

3

[3 marks]

A radioactive substance decays by 8% each day.

There are 400 g at the start.

Work out the mass after 10 days, to 1 decimal place.

1 The multiplier for an 8% decrease is 0.92

2 $\text{Mass} = 400 \times 0.92^{10}$

3 $0.92^{10} = 0.4343884\dots$

4 $400 \times 0.4343884 = 173.755\dots \text{ g}$

Answer: 173.8 g (1 d.p.)

4

[3 marks]

An investment of £4000 grows to £4630.50 in 3 years with compound interest.

Work out the annual interest rate.

1 The total multiplier over 3 years is $\frac{4630.50}{4000} = 1.157625$

2 The yearly multiplier is the cube root of this.

3 The cube root of 1.157625 is 1.05 (since $1.05^3 = 1.157625$).

4 A multiplier of 1.05 means a 5% increase.

Answer: 5% per year

5

[4 marks]

£6000 is invested at 3% compound interest per year.

Work out the least number of complete years before the investment exceeds £8000.

1 Each year the value is multiplied by 1.03

2 After 9 years: $6000 \times 1.03^9 = 6000 \times 1.304773 = £7828.64$ — not enough.

3 After 10 years: $6000 \times 1.03^{10} = 6000 \times 1.343916 = £8063.50$

4 This is the first year the value is above £8000.

Answer: 10 years

6

[4 marks]

A machine is worth £40 000 when new.

It loses 25% of its value in the first year and 15% of its value in each year after that.

Work out its value after 4 years.

1 First year multiplier: 0.75. Later years multiplier: 0.85, used three times.

2 $\text{Value} = 40\,000 \times 0.75 \times 0.85^3$

3 $40\,000 \times 0.75 = £30\,000$

4 $0.85^3 = 0.614125$

5 $30\,000 \times 0.614125 = £18\,423.75$

Answer: £18 423.75

7

[4 marks]

The number of users N of an app after t months is modelled by

$$N = 500 \times 1.4^t$$

- (a) How many users were there at the start?
 (b) How many users after 6 months? Give your answer to the nearest whole number.
 (c) What does the number 1.4 tell you?

1 (a) At $t = 0$, $1.4^0 = 1$, so $N = 500$ users.

2 (b) $1.4^6 = 7.529536$

3 $N = 500 \times 7.529536 = 3764.768$, so about 3765 users.

4 (c) Each month the number of users is multiplied by 1.4, which is a 40% increase per month.

Answer: (a) 500 (b) 3765 (c) the number of users grows by 40% each month

8

[4 marks]

A substance has a half-life of 12 years, so its mass halves every 12 years.

There are 640 g at the start.

- (a) Work out the mass after 48 years.
 (b) Write a formula for the mass m grams after t years.

1 (a) $48 \div 12 = 4$, so the mass halves 4 times.

2 $640 \rightarrow 320 \rightarrow 160 \rightarrow 80 \rightarrow 40$

3 The mass is 40 g. (Or $640 \div 2^4 = 640 \div 16 = 40$.)

4 (b) The number of half-lives after t years is $\frac{t}{12}$

5 So $m = 640 \times \left(\frac{1}{2}\right)^{t/12}$

Answer: (a) 40 g (b) $m = 640 \times \left(\frac{1}{2}\right)^{t/12}$

9

[4 marks]

£5000 is invested for one year. Two options are offered.

Option 1: 6% interest added once, at the end of the year.

Option 2: 0.5% interest added at the end of every month.

Which option gives more money, and by how much?

1 **Option 1:** $5000 \times 1.06 = £5300$

2 **Option 2:** the multiplier 1.005 is applied 12 times.

3 $1.005^{12} = 1.0616778\dots$

4 $5000 \times 1.0616778 = £5308.39$

5 $5308.39 - 5300 = £8.39$

6 Option 2 is better because the later months earn interest on interest already added.

Answer: Option 2, by £8.39

An amount £P is invested at 4% compound interest per year.

After 5 years it is worth £9733.22

(a) Work out P.

(b) Work out how many more complete years it takes for the investment to exceed £12 000.

1 (a) The multiplier over 5 years is $1.04^5 = 1.2166529$

2 $P \times 1.2166529 = 9733.22$, so $P = 9733.22 \div 1.2166529 = £8000$

3 (b) Start from £9733.22 and keep multiplying by 1.04

4 After 5 more years: $9733.22 \times 1.2166529 = £11\,841.72$ — not yet enough.

5 After 6 more years: $9733.22 \times 1.04^6 = 9733.22 \times 1.2653190 = £12\,315.39$

6 So it takes 6 more complete years.

Answer: (a) £8000 (b) 6 more years